ACC250: Intro to Financial Accounting

Ch9. Long-lived assets

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Learning Objectives

- Classify assets as **tangible** or **intangible**.
- Determine **acquisition cost** under the **cost principle**.
- Depreciate assets by the **straight-line** and **double-declining-balance** methods.
- Track **accumulated depreciation** and **book value**.

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2. Tangible assets - Introduction
3. Tangible assets - Acquisition
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Assets

Assets (A)

Resources owned by a company that are expected to provide future economic benefits.

Characteristics of assets:

- Owned or controlled by the business
- Expected to provide _____
- Measurable in monetary terms

Carrying Value or Book Value

The value of an asset reported on the balance sheet.

- BV is initially equal to the acquisition cost under _____ principle.
- BV is adjusted for depreciation, amortization, and impairment, depending on the asset type.
- BV is not necessarily equal to the _____ of the asset.
- BV indicates _____ BV unless it is specified for _____ BV.

EX. For PPE:

- Gross BV is $1M
- Accumulated Depreciation is $0.3M
- $\rightarrow$ Net BV is _____.

Assets - Classification

Two important questions for classification:

1. How long will the asset contribute to revenue generation?
2. Is the asset physical and visible?

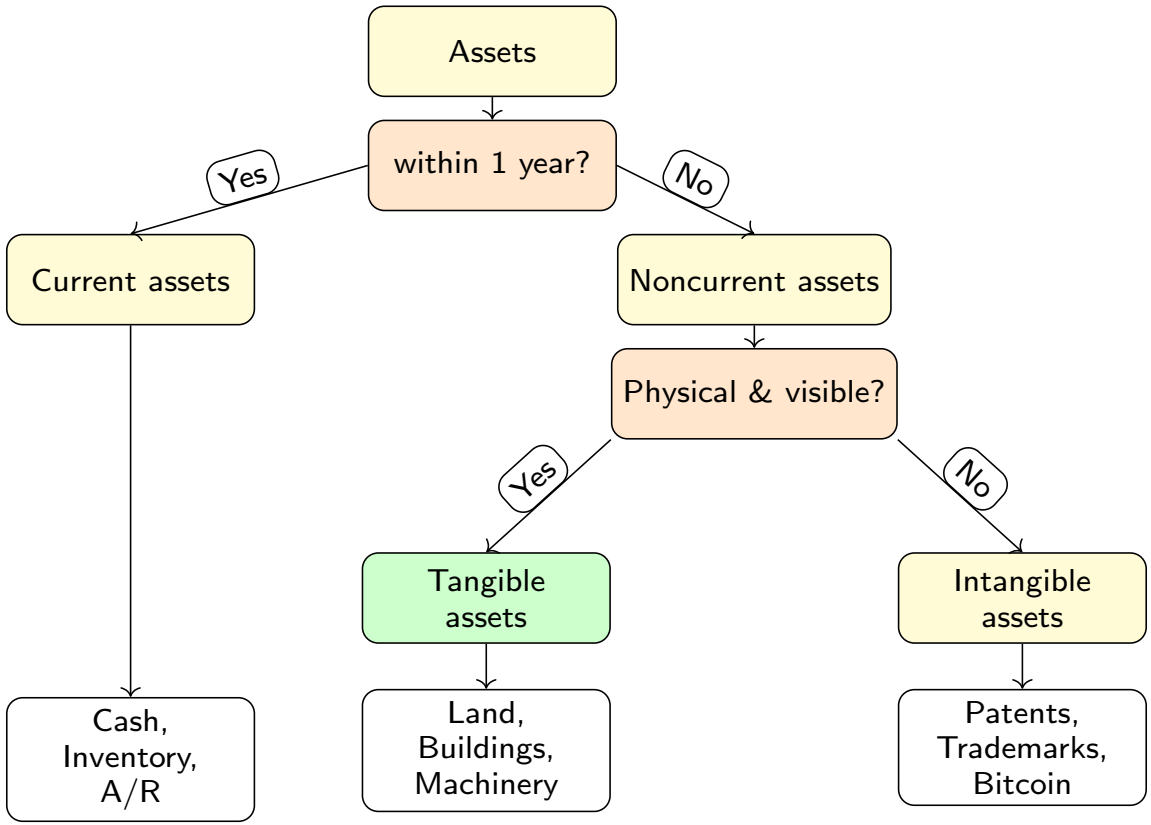


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Tangible assets

Tangible assets

Physical assets that are expected to provide future economic benefits for more than one year.

- Land
- Buildings
- Equipment
- Property

Tangible assets

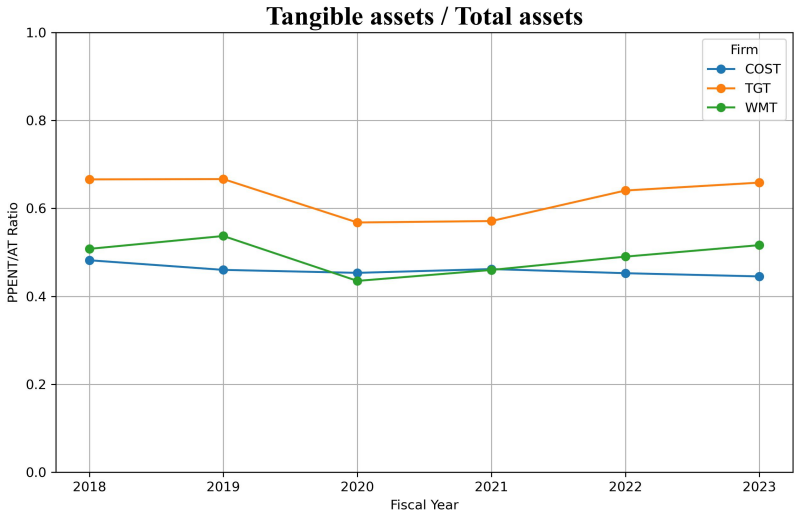
Property, Plant, and Equipment (PPE, Assets)

Long-term tangible assets that a company uses to operate its business, such as buildings, machinery, and vehicles, with a useful life of more than one year.





How significant are **tangible assets relative to total assets** in the balance sheet?



Note: COST: Costco; TGT: Target; WMT: Walmart.

Tangible assets: Disclosures

PPE and Accumulated Depreciation

Balance Sheet
As of Dec 31, 2025

Assets	
.....	...
.....	...
PPE (at cost)	$100,000
Less: Acc.Dep.Exp	($40,000)
_____	**$60,000**
.....	...
Total Assets	$200,000

Depreciation Expense

Income Statement
For the Year Ended Dec 31, 2025

Revenues	$220,000
Cost of Goods Sold	($120,000)
Gross Profit	$100,000
_____	($20,000)
Net Income	**$80,000**

PPE, DepExp and AccDep are inter-related!

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Tangible assets: Acquisition

Cost Principle

Assets are initially recorded at their **acquisition cost**, which includes all costs necessary to acquire the asset and prepare it for its intended use.

Acquisition cost is the total cost incurred to acquire an asset and prepare it for its intended use. It includes:

- Purchase price
- Sales tax
- Delivery and installation costs
- Legal fees
- Discounts (deduction)

EX. ABC paid $100K for a machine. Additional costs (all paid by cash) include Sales tax: $10K; Delivery: $5K; and Installation: $3K. How much to recognize as the acquisition cost of the machine?

Tangible assets: Acquisition

Note that expenditures such as sales tax and delivery costs are _____ as part of the asset's acquisition cost.

Capitalization

The process of recording an expenditure as an asset rather than an expense.

- Capitalized costs are added to the asset's acquisition cost.
- So it's part of the asset's value on the balance sheet.
- _____ immediately on the income statement.
- Instead, the costs are _____ over the asset's useful life.

EX. On Jan 1, 2026, A company purchases machinery for $50K by cash.

Questions:

1. Journal entries on Jan 1, 2026:

EX. On Jan 1, 2026, A company purchases machinery for $50K by cash. Additional costs (all paid by cash) include Sales tax: $5K; Delivery: $3K; and Installation: $2K.

Questions:

1. Total acquisition cost:

2. Journal entries on Jan 1, 2026:

EX. On Jan 1, 2026, A company purchases machinery for $50K less a $10K discount. The company signed a note payable for the machinery.

Questions:

1. Total acquisition cost:

2. Journal entries on Jan 1, 2026:

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Tangible assets: Depreciation

Q. Companies pay cash to buy PPE. But no expenses?

Matching Principle

- Expenses should be recognized **in the same period as the revenues** they help to generate, regardless of when cash is paid. In other words,
- Expenses should be recognized when the resources are **used up** to generate the revenues.

- Long-term assets contribute the revenue generation over multiple periods.
- So it is _____ in the period of acquisition.
- Instead, the cost is _____ over the asset's useful life.
- The allocation process is called "_____".

Tangible assets: Depreciation

Depreciation

The process of allocating the cost of a long-term, tangible asset over its useful life.

- Depreciation expense is recorded on the income statement.
- Accumulated depreciation is recorded on the balance sheet as a _____.

Tangible assets: Depreciation

Depreciation Expense (DepExp, Expense)

The amount of the tangible asset's cost that is allocated to the current accounting period.

Accumulated Depreciation (AccDep, Contra Asset, xA)

The total amount of depreciation expense that has been recorded against the asset since it was acquired.

Tangible assets: Depreciation

Q1. Does the **recognition of depreciation** come with a **cash outflow**?

1. _____. We pay cash when we acquire the asset.
2. The recognition of depreciation is just _____.[1]

Q2. Does the **depreciation process** have something to do with the asset's **market value**?

1. Market value is how much the asset is worth in the market.
2. Depreciation expense and Net BV depends on accounting methods, not by how much we would get if we sold the asset.
3. So it is "_____".

[1] Deferral adjustments in Ch4.

Tangible assets: Depreciation

Two widely used methods of depreciation:

1. Straight-line method (**SLM**)
2. Double-declining balance method (**DDM**)

Tangible assets: Depreciation - Straight-line method (SLM)

Straight-line method

- The most common method of depreciation.
- Allocates an equal amount of depreciation expense to each accounting period over the asset's useful life.
- $$\text{Annual DepExp} = \frac{\text{Cost} - \text{Residual Value}}{\text{Useful Life}}$$

Cost - The total acquisition cost of the asset.
Residual value - The estimated value of the asset at the end of its useful life.
Useful life - The estimated period over which the asset will be used.

EX. A company purchases equipment on Jan 1, 2026, for \$60K. The equipment has a residual value of \$0 and a useful life of 5 years. Calculate the annual depreciation expense.

1. Annual depreciation expense:

2. Journal entries on Dec 31, 2026:

3. Journal entries on Dec 31, 2027:

4. 5 years later, the expected value of the asset?:

Year	Cost Value	Beg BV	DepExp	AccDep	Ending BV
1	$60,000	$60,000	$12,000	$12,000	$48,000
2	$60,000	$48,000	$12,000	$24,000	$36,000
3	$60,000	$36,000	$12,000	$36,000	$24,000
4	$60,000	$24,000	$12,000	$48,000	$12,000
5	$60,000	$12,000	$12,000	$60,000	$0

Table: Depreciation Schedule for Straight-Line Method

Beg BV= Last year's Ending BV.
Ending BV= _____.

EX. A company purchases equipment on Jan 1, 2026, for \$40K. The equipment has a residual value of \$4K and a useful life of 4 years. Calculate the annual depreciation expense.

1. Annual depreciation expense:

2. Journal entries on Dec 31, 2026:

3. Journal entries on Dec 31, 2027:

4. 4 years later, the expected value of the asset?

5. This asset, if sold, will generate \$4K in cash flows.

Year	Cost Value	Beg BV	DepExp	AccDep	Ending BV
1	$40,000	$40,000	$9,000	$9,000	$31,000
2	$40,000	$31,000	$9,000	$18,000	$22,000
3	$40,000	$22,000	$9,000	$27,000	$13,000
4	$40,000	$13,000	$9,000	$36,000	$4,000

Table: Depreciation Schedule for Straight-Line Method

$$\text{Beg BV} = \text{Last year's Ending BV}$$
$$\text{Ending BV} = \text{Cost} - \text{AccDep}$$

Tangible assets: Depreciation - Double-declining (DDM)

Double-declining balance method

- Allocates a _____ depreciation expense in the earlier years of the asset's useful life and a lower expense in the later years.
- $$\text{Annual DepExp} = \frac{2}{\text{Useful Life}} \times (\text{Cost - AccDep})$$

  Cost - The total acquisition cost of the asset.
  AccDep - The AccDep at the beginning of the year.
  Useful life - The estimated period over which the asset will be used.
  Annual DepExp - The annual DepExp for the current year.
- The above formula applies to all years but the last year.
- **Special treatment for the last year:**
  - The last DepExp is **obtained** as the amount that brings the Ending BV down to the residual value.[2]

[2] It'll be much clearer through the following example.

Tangible assets: Depreciation - Double-declining

Double-declining balance method

- Annual DepExp $= \frac{2}{\text{Useful Life}} \times$ (Cost - AccDep)
- AccDep _____ over time.
- (Cost - AccDep) _____ over time.
- DepExp _____ over time.

Straight-line method

- Annual DepExp $= \frac{\text{Cost} - \text{Residual Value}}{\text{Useful Life}}$
- DecExp is _____ over time.

EX. A company purchases equipment on Jan 1, 2026, for \$40K. The equipment has a residual value of \$4K and a useful life of 4 years. Calculate the depreciation expense for Year 1 through Year 4 using the double-declining balance method.

1. Depreciation rate:

2. Year 1 depreciation expense:

3. Year 2 depreciation expense:

4. Year 3 depreciation expense:

5. Year 4 depreciation expense (the **last** year):

Year	Cost Value	Beg BV	DepExp	AccDep	Ending BV
1	$40,000	$40,000	$20,000	$20,000	$20,000
2	$40,000	$20,000	$10,000	$30,000	$10,000
3	$40,000	$10,000	$5,000	$35,000	$5,000
4	$40,000	$5,000	$1,000	$36,000	$4,000

Table: Depreciation Schedule for Double-Declining Balance Method

$$\text{Beg BV} = \text{Last year's Ending BV}$$
$$\text{Ending BV} = \text{Cost} - \text{AccDep}$$

Comparison with SLM

EX. The **same** equipment: cost \$40K, residual value \$4K, useful life 4 years.

Straight-line: Annual DepExp $= \frac{\$40K - \$4K}{4} =$ ____ every year.

Year	SLM DepExp	DDM DepExp
1	\$9,000	\$20,000
2	\$9,000	\$10,000
3	\$9,000	\$5,000
4	\$9,000	\$1,000
Total	**\$36,000**	**\$36,000**

- Both methods depreciate the **same total** (\$36,000) and end at the residual value (\$4,000).
- DDM recognizes _____ expense in the early years and _____ in the later years.